

One Learning Rate Is Not Enough

Every parameter gets its own step size, measured from its own gradients. Fill every blank by clicking. Nothing here is scored.

START HERE

How to use this sheet

- Click the option you think is right. It drops into the blank.
- Wrong answers grey out and give you a hint. Nothing is scored.
- Each step unlocks when the one before it is filled. Everything else stays readable.
- Every calculation gives you the numbers it needs.

PART 1

One step size, two very different parameters

THE QUESTION

Momentum changed the direction we walk in. It did not change how far. What happens when two parameters need steps of completely different sizes?

Two parameters. One α . Nothing else is new.

The setup

Two parameters of the same network. w_1 sits on a steep direction of the loss, w_2 on a flat one. Both start at 1.0000. Here are their gradients, as measured:

TABLE A · THE GRADIENTS, AS MEASURED

	step 1	step 2	what it is
g_t for w_1	3	4	a steep direction
g_t for w_2	0.3	0.4	a flat direction, ten times smaller every step

The update is the same for both: $w_t = w_{t-1} - \alpha g_t$, with $\alpha = 0.1$.

TWO STEPS OF PLAIN SGD

w_1 : $1.0000 - 0.1(3) = \mathbf{0.7000}$, then $0.7000 - 0.1(4) = \mathbf{0.3000}$

w_2 : $1.0000 - 0.1(0.3) = \mathbf{0.9700}$, then $0.9700 - 0.1(0.4) = \mathbf{0.9300}$

w_1 moved 0.7000. w_2 moved 0.0700. Ten times apart, because the step is α times the gradient and nothing else.

Lower α and w_1 behaves, but w_2 stops moving. Raise it and w_2 moves, but w_1 overshoots. Run both cases.

Animation 1 · one α , two directions

STEP 120 W1 0.0000 W2 3.8142 STEPS FOR W2 TO FINISH **574 steps**

What the animation measures

α	steps for the steep direction	steps for the flat direction	with momentum, $\beta = 0.9$
0.02	21	2301	2268
0.08	3	574	539
0.19	44	241	200
0.22	diverges		165

At $\alpha = 0.08$ the steep direction is finished in 3 steps and the flat one needs 574. Momentum brings that to 539, a 6% improvement. It is not the fix.

PART 1 · QUESTION 1

What would let both parameters move at a sensible rate at the same time?

- A A smaller α .
- B Momentum with a larger β .
- C A separate step size for each parameter, set from that parameter's own gradients.
- D A better starting point.

THE ANSWER

Stop using the gradient's size to set the distance. We cannot know the right step size for each parameter in advance. We can measure it.

A parameter whose gradients have been consistently large is on a steep direction and should take smaller steps. A parameter whose gradients have been consistently small is on a flat direction and should take larger ones. So **divide each parameter's step by a running measure of that parameter's own gradient size.**

Everything in the rest of this sheet is one question: what should that running measure be?

PART 2

AdaGrad: divide by your own history

The simplest running measure of a parameter's gradient size: add up the squares. Squares, so that sign never cancels.

ADAGRAD · ONE G PER PARAMETER

$G_t = G_{t-1} + g^2_t$

$w_t = w_{t-1} - \alpha g_t / (\sqrt{G_t} + \epsilon)$

$G_0 = 0$. Each parameter keeps its own G , so each parameter gets its own denominator.

ϵ is about 10^{-8} . It is there so the first division cannot be by zero, and it changes nothing at 4 decimals. We drop it from the arithmetic from here on.

The two gradients from Part 1, through the new rule

STEP 1

$w_1:$ $G_1 = 0 + 3^2 = \mathbf{9.0000}$, $\sqrt{9.0000} = 3.0000$, $\text{step} = 0.1(3)/3.0000 = \mathbf{0.1000}$

$w_2:$ $G_1 = 0 + 0.3^2 = \mathbf{0.0900}$, $\sqrt{0.0900} = 0.3000$, $\text{step} = 0.1(0.3)/0.3000 = \mathbf{0.1000}$

STEP 2

$w_1:$ $G_2 = 9.0000 + 4^2 = \mathbf{25.0000}$, $\sqrt{25.0000} = 5.0000$, $\text{step} = 0.1(4)/5.0000 = \mathbf{0.0800}$

$w_2:$ $G_2 = 0.0900 + 0.4^2 = \mathbf{0.2500}$, $\sqrt{0.2500} = 0.5000$, $\text{step} = 0.1(0.4)/0.5000 = \mathbf{0.0800}$

The gradients are ten times apart. The steps are equal. Multiplying every gradient of a parameter by 10 multiplies $\sqrt{G}$ by 10 as well, and the ratio $g/\sqrt{G}$ does not move.

WHAT ADAGRAD CHANGED

The gradient now sets the direction of the step. It no longer sets the distance.

At step 1, $G_1 = g^2_1$, so the step is $\alpha g_1/|g_1| = \alpha$ exactly, whatever the gradient was. Every AdaGrad run starts with a step of exactly α .

PART 2 · YOUR TURN

AdaGrad by hand

THE RULE, LEARNING RATE 0.1

$G_t = G_{t-1} + g^2_t \quad \cdot \quad w_t = w_{t-1} - \alpha g_t / \sqrt{G_t}$

What you are given

$w_1 = w_2 = 1.0000$

$\alpha = 0.1$

$G_0 = 0$ for both

gradients for w_1 : 3, then 4

gradients for w_2 : 0.3, then 0.4



STEP 1
both start at 1.0000
watch how far each one moves.

YOU NEED $G_0 = 0$ $g_1 = 3$

$G_1 = 0 + 3^2 =$ _____

- (a) 9.0000
- (b) 0.0900
- (c) 6.0000
- (d) 3.0000

YOU NEED $G_1 = 9.0000$ $\alpha = 0.1$

$\Delta w_1 = 0.1(3) / \sqrt{9.0000} = 0.1(3) /$ _____ $=$ _____

THE SQUARE ROOT OF THE ACCUMULATOR

- (a) 81.0000
- (b) 4.5000
- (c) 3.0000
- (d) 9.0000

THE STEP FOR w_1

- (a) 0.3000
- (b) 3.0000
- (c) 0.0333
- (d) 0.1000

YOU NEED g_1 for $w_2 = 0.3$

Now w_2 , whose gradient is 0.3. $G_1 = 0.3^2 =$ _____, whose square root is 0.3000.

Then $\Delta w_2 = 0.1(0.3) / 0.3000 =$ _____

THE ACCUMULATOR FOR w_2

- (a) 0.0900
- (b) 0.3000
- (c) 0.6000
- (d) 9.0000

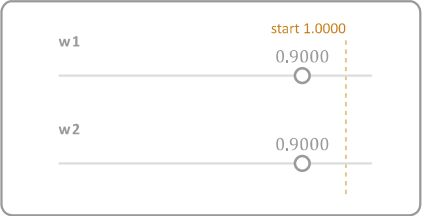
THE STEP FOR w_2

- (a) 0.1000
- (b) 0.0300
- (c) 1.0000
- (d) 0.0100

YOU NEED step for $w_1 = 0.1000$ step for $w_2 = 0.1000$

The two gradients differ by a factor of 10. The two steps differ by a factor of

- _____
- (a) 10
 - (b) 1, they are identical
 - (c) 100
 - (d) 3.16



STEP 2
both at **0.9000**
the accumulator has grown, so
the step shrinks.

YOU NEED $G_1 = 9.0000$ $g_2 = 4$

$G_2 = 9.0000 + 4^2 =$ _____

- (a) 16.0000
- (b) 25.0000
- (c) 5.0000
- (d) 13.0000

YOU NEED $G_2 = 25.0000$ $\alpha = 0.1$

$\Delta w_1 = 0.1(4) / \sqrt{25.0000} = 0.1(4) /$ _____ $=$ _____

THE SQUARE ROOT OF THE ACCUMULATOR

- (a) 5.0000
- (b) 12.5000
- (c) 25.0000
- (d) 4.0000

THE STEP FOR w_1

- (a) 0.4000
- (b) 0.0500
- (c) 0.1000
- (d) 0.0800

YOU NEED w_1 after step 1 = 0.9000 step = 0.0800

$w_1 = 0.9000 - 0.0800 =$ _____

- (a) 0.8000
- (b) 0.9800
- (c) 1.0800
- (d) 0.8200

YOU NEED $w_1 = 0.8200$ both steps were equal

w_2 's accumulator is 0.2500 and its step is also 0.0800. After two steps AdaGrad leaves
the two parameters at _____

- (a) 0.8200 and 0.9300
- (b) 0.3000 and 0.9300
- (c) 0.9000 and 0.9000
- (d) 0.8200 and 0.8200

Two parameters, gradients ten times apart, identical progress

TABLE B · ADAGRAD, LEARNING RATE 0.1

	G_1	step 1	G_2	step 2	w after
w_1	9.0000	0.1000	25.0000	0.0800	0.8200
w_2	0.0900	0.1000	0.2500	0.0800	0.8200

Plain SGD left them at 0.3000 and 0.9300. AdaGrad left them at 0.8200 and 0.8200.

PART 3

What AdaGrad breaks

THE QUESTION

g^2_t is never negative, so G can only grow. What is left of α after a thousand steps?

Look at the denominator, not the numerator.

Animation 2 · the step size, over time

STEPS SHOWN 1000 EFFECTIVE STEP AT THE END 0.00491 as a fraction of α 0.049 $\times \alpha$ α halves at step 5 W2 STILL TO GO 4.625

TABLE C · A STEADY GRADIENT: WHAT IS LEFT OF THE LEARNING RATE

after t steps	G_t	effective step, $\alpha / \sqrt{G_t}$
1	$t g^2$	$1.0000 \alpha / g$
4	$t g^2$	$0.5000 \alpha / g$
25	$t g^2$	$0.2000 \alpha / g$
100	$t g^2$	$0.1000 \alpha / g$
400	$t g^2$	$0.0500 \alpha / g$
2500	$t g^2$	$0.0200 \alpha / g$
10000	$t g^2$	$0.0100 \alpha / g$

Measured on the Part 1 loss, $\alpha = 0.1$

after t steps	effective step for the flat direction	as a fraction of α
1	0.1000	1.000
10	0.0325	0.325
100	0.0113	0.113
1000	0.0049	0.049

AdaGrad needs 10590 steps to bring the flat direction within 1% of zero. Plain SGD at $\alpha = 0.19$ needed 241. Equalising the two parameters cost us the ability to finish.

PART 3 · QUESTION 2

Why does AdaGrad stall before it converges?

A

The gradients have gone to zero.

B

Nothing in the rule can ever make G smaller, so the denominator grows without limit.

C

α was chosen too small.

D

The two parameters interfere with each other.

THE ANSWER

The measure has no way to forget. A gradient from step 5 counts as much in G as a gradient from step 5000, even though the parameter has moved a long way since.

We want the same normalising division, but from a measure that tracks the recent gradient size rather than all of history. You built exactly that tool in Worksheet 07.

PART 4

RMSProp: an average that forgets

Worksheet 07 ran an exponentially weighted moving average over gradients to smooth the direction. Run the same average over **squared** gradients to track the size.

RMSPROP

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g^2_t$$
$$w_t = w_{t-1} - \alpha g_t / (\sqrt{v_t} + \epsilon)$$

Only the accumulator changed. The update line is AdaGrad's, with v in place of G .

THE WHOLE DIFFERENCE

AdaGrad adds every squared gradient. RMSProp averages the recent ones. β_2 decides how many count as recent: about $1/(1 - \beta_2)$ of them.

Animation 3 · what β_2 buys

STEP 128

β_2 0.9

REMEMBERS ABOUT 10 gradients

RMSPROP STEP 0.0200

ADAGRAD STEP 0.0024

STEPS TO REACT TO THE JUMP 18 steps

The same two gradients again, $\beta_2 = 0.9$

w_1 , BOTH STEPS

$$v_1 = 0.9(0) + 0.1(9.0000) = \mathbf{0.9000}, \quad \sqrt{0.9000} = 0.9487, \quad \text{step} = 0.1(3)/0.9487 = \mathbf{0.3162}$$
$$v_2 = 0.9(0.9000) + 0.1(16.0000) = 0.8100 + 1.6000 = \mathbf{2.4100}, \quad \sqrt{2.4100} = 1.5524, \quad \text{step} = 0.1(4)/1.5524 = \mathbf{0.2577}$$

w_2 takes exactly the same two steps, for the same reason as in Part 2: scaling every gradient of a parameter scales $\sqrt{v}$ with it.

TABLE D · RMSPROP, LEARNING RATE 0.1, DECAY 0.9

	v_1	step 1	v_2	step 2	w after
w_1	0.9000	0.3162	2.4100	0.2577	0.4261
w_2	0.0090	0.3162	0.0241	0.2577	0.4261

β_2 is usually 0.9 to 0.999. This sheet uses 0.9 so that two steps of arithmetic show you something.

PART 4 · YOUR TURN

RMSPprop by hand

THE RULE, LEARNING RATE 0.1, DECAY 0.9

$$v_t = 0.9 v_{t-1} + 0.1 g^2_t \quad \cdot \quad w_t = w_{t-1} - 0.1 g_t / \sqrt{v_t}$$

What you are given

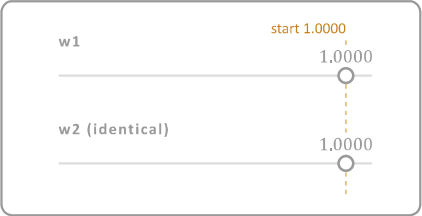
$w_1 = 1.0000$

$v_0 = 0$

$\alpha = 0.1, \beta_2 = 0.9$

gradients: 3, then 4 · squared: 9.0000, then 16.0000

Only w_1 this time. w_2 takes the identical steps.



STEP 1
from 1.0000
 v starts at zero. Notice how far this step goes.

YOU NEED $v_0 = 0$ $\beta_2 = 0.9$ $g^2_1 = 9.0000$

$v_1 = 0.9(0) + 0.1(9.0000) =$ _____

- (a) 0.0900
- (b) 8.1000
- (c) 9.0000
- (d) 0.9000

YOU NEED $v_1 = 0.9000$ $\alpha = 0.1$

$\sqrt{0.9000} =$ _____, then $\Delta w_1 = 0.1(3) \div$ that = _____

THE SQUARE ROOT

- (a) 0.4500
- (b) 0.8100
- (c) 0.3000
- (d) 0.9487

THE STEP FOR w_1

- (a) 0.3162
- (b) 0.0949
- (c) 0.1000
- (d) 0.3000

YOU NEED $w_0 = 1.0000$ step = 0.3162

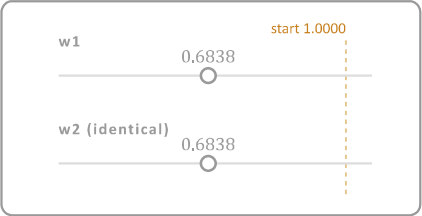
$w_1 = 1.0000 - 0.3162 =$ _____

- (a) 0.7000
- (b) 1.3162
- (c) 0.6838
- (d) 0.9684

YOU NEED $\alpha = 0.1$ step = 0.3162 $\beta_2 = 0.9$

α is 0.1000. The step you just took is 0.3162, which is _____

- (a) the same as α
- (b) 0.9 times α
- (c) 3.16 times α
- (d) a tenth of α



STEP 2
from **0.6838**
v now has a real value in it.

YOU NEED $v_1 = 0.9000$ $g^2_2 = 16.0000$ $\beta_2 = 0.9$

$v_2 = 0.9(0.9000) + 0.1(16.0000) = 0.8100 + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

- THE NEW SQUARED GRADIENT, WEIGHTED
- (a) 16.0000
 - (b) 1.6000
 - (c) 14.4000
 - (d) 0.1600

- THE NEW AVERAGE
- (a) 2.4100
 - (b) 16.9000
 - (c) 0.8100
 - (d) 1.6000

YOU NEED $v_2 = 2.4100$ $\alpha = 0.1$

$\sqrt{2.4100} = \underline{\hspace{1cm}}$, then $\Delta w_1 = 0.1(4) \div \text{that} = \underline{\hspace{1cm}}$

- THE SQUARE ROOT
- (a) 1.2050
 - (b) 1.5524
 - (c) 5.8081
 - (d) 2.4100

- THE STEP FOR w_1
- (a) 0.3162
 - (b) 0.2577
 - (c) 0.0800
 - (d) 0.4000

YOU NEED w_1 after step 1 = 0.6838 $\text{step} = 0.2577$

$w_1 = 0.6838 - 0.2577 = \underline{\hspace{1cm}}$

- (a) 0.3000
- (b) 0.9415
- (c) 0.4261
- (d) 0.5261

You found RMSProp's own problem with your own hands

$v_0 = 0$, so $v_1 = (1 - \beta_2)g^2_1$. That is not an estimate of g^2_1 : it is one tenth of it. The square root is 3.16 times too small, so the first step is 3.16 times too large.

first step = $\alpha g_1 / \sqrt{((1 - \beta_2)g^2_1)} = \alpha / \sqrt{(1 - \beta_2)}$

The gradient cancels. The size of the error depends only on β_2 .

TABLE E · RMSPROP'S FIRST STEP, IN UNITS OF THE LEARNING RATE

β_2	$1 - \beta_2$	first step, in units of α
0.5	0.500	1.41 ×
0.9	0.100	3.16 ×
0.99	0.010	10.00 ×
0.999	0.001	31.62 ×

At the common default $\beta_2 = 0.999$, RMSProp's first step is 31.6 times the size you asked for.

PART 5

What RMSProp still gets wrong

THE QUESTION

Two things are still wrong. One you just measured. What is the other?

Look at the numerator of the RMSProp update, and compare it with Worksheet 07.

The two gaps

1. The average starts at zero.

You measured it: the first step is $\alpha/\sqrt{1 - \beta_2}$, which at $\beta_2 = 0.999$ is 31.6 times too large.

2. The numerator is a single raw gradient.

Worksheet 07 showed that one gradient is a noisy estimate of the direction. RMSProp fixes the size of the step and then points it with a raw gradient anyway.

You already have both halves

Momentum averaged the gradient. RMSProp averaged the squared gradient. Two averages, two different jobs.

THE TWO AVERAGES

DIRECTION

$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$

which way have the gradients been pointing?

SIZE

$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g^2_t$

how large are this parameter's gradients, usually?

Why one of them squares and the other does not

Take four gradients that keep reversing: **+3, -3, +3, -3**.

	running total	what it says
average of g	0	no reliable direction, which is correct
average of g^2	36	the gradients were large, which is also correct

The cancellation is the point. m must keep the sign, because a direction that keeps reversing is not worth trusting. v must throw the sign away, because -10 is just as large a gradient as $+10$.

PART 5 · QUESTION 3

A parameter's gradients are **+2, -2, +2, -2, over and over**. What do the two averages settle near?

A

m near 4, v near 0

B

m near 0, v near 4

C

both near 0

D

both near 4

SO FAR

Keep both averages. Use m for the direction and $\sqrt{v}$ for the distance. One problem is left: **both of them start at zero**.

PART 6

Where the correction comes from

THE QUESTION

Hold the gradient fixed at 1, for ever. What does the moving average report?

It should report 1. Work out the first three values and see.

A constant gradient, $\beta = 0.9$, $m_0 = 0$

$m_t = 0.9\ m_{t-1} + 0.1\ g_t$ with $g_t = 1$ at every step

Every gradient is 1, so the honest average of the gradients is 1. Watch what the rule actually returns.

YOU NEED

$m_0 = 0$

$\beta = 0.9$

$g_1 = 1$

$m_1 = 0.9(0) + 0.1(1) =$ _____

(a) 0.0100

(b) 0.9000

(c) 1.0000

(d) 0.1000

YOU NEED

$m_1 = 0.1000$

$\beta = 0.9$

$g_2 = 1$

$m_2 = 0.9(0.1000) + 0.1(1) =$ _____ $+ 0.1000 =$ _____

THE OLD AVERAGE, WEIGHTED

- (a) 0.0900

(b) 0.1000

(c) 0.0090

(d) 0.9000

THE NEW AVERAGE

- (a) 0.1000

(b) 0.2000

(c) 1.0000

(d) 0.1900

YOU NEED

$m_2 = 0.1900$

$\beta = 0.9$

$g_3 = 1$

$m_3 = 0.9(0.1900) + 0.1(1) = 0.1710 + 0.1000 =$ _____

(a) 0.2710

(b) 0.1900

(c) 0.3710

(d) 1.0000

YOU NEED

$m_1 = 0.1000$

$m_2 = 0.1900$

$m_3 = 0.2710$

Every gradient was 1, so the honest average is 1.0000. After three steps the rule reports _____

(a) 1.0000, exactly right

(b) 3.0000

(c) 0.2710, about a quarter of it

(d) 0.9000

The average is not wrong. It is incomplete.

Nothing has gone wrong with the arithmetic. The rule has simply not had time to shift its weight off the zero it started from.

$m_3 = 0.1(1) + 0.09(1) + 0.081(1) + 0.729(0) = 0.2710$

Three real observations carry 0.271 of the weight. The remaining **0.729** is still sitting on $m_0 = 0$.

YOU NEED

0.1000

0.1900

0.2710

Compare your three answers with $1 - 0.9^1 = 0.1$, $1 - 0.9^2 = 0.19$, $1 - 0.9^3 = 0.271$. They are

-
- (a) the reciprocals
- (b) unrelated
- (c) close, but not equal
- (d) the same three numbers

Animation 4 · the weight that is missing

STEPS SO FAR 3 $\beta = 0.9$ weight applied, $1 - \beta^t$ 0.2710 weight still on zero, β^t 0.7290 THE CORRECTION DIVIDES BY THIS 3.6900 ×

The pattern

TABLE F · A CONSTANT GRADIENT: WHAT THE AVERAGE REPORTS, AS A FRACTION OF IT

after t steps	$\beta = 0.9$	$\beta = 0.999$	how far off
1	0.1000	0.0010	0.9 is 90% short, 0.999 is 100% short
2	0.1900	0.0020	0.9 is 81% short, 0.999 is 100% short
3	0.2710	0.0030	0.9 is 73% short, 0.999 is 100% short
10	0.6513	0.0100	0.9 is 35% short, 0.999 is 99% short
50	0.9948	0.0488	0.9 is 1% short, 0.999 is 95% short
300	1.0000	0.2593	0.9 has arrived, 0.999 is 74% short
3000	1.0000	0.9503	0.9 has arrived, 0.999 is 5% short

0.1000, 0.1900, 0.2710 are exactly $1 - 0.9^1$, $1 - 0.9^2$, $1 - 0.9^3$. For a constant gradient g , the rule returns $(1 - \beta^t)g$, never g .

YOU NEED

$0.9^3 = 0.7290$

$m_3 = 0.2710$

The weight actually applied after 3 steps is $1 - 0.9^3 = 1 - 0.7290 =$ _____

-
- (a) 0.9000
- (b) 0.7290
- (c) 0.2710
- (d) 1.7290

THE CORRECTION

The running total is short by exactly the factor $1 - \beta^t$. So divide by it.

$\hat{m}_t = m_t / (1 - \beta_1^t) \quad \cdot \quad \hat{v}_t = v_t / (1 - \beta_2^t)$

check: $0.2710 / (1 - 0.9^3) = 0.2710 / 0.2710 = 1.0000$

Dividing a weighted sum by the total weight is what makes it an average. Nothing more is happening here.

The correction switches itself off. As t grows, β^t goes to zero, $1 - \beta^t$ goes to 1, and dividing by 1 changes nothing.

What it costs to skip it

Animation 5 · correction on, correction off

β_2 0.999 FIRST STEP 3.16 × α FURTHEST FROM CORRECT 6.57 × α WHICH HAPPENS AT step 12 WITHIN 5% OF CORRECT FROM **step 2375**

What the animation measures

For a steady gradient the uncorrected step is the correct one multiplied by $(1 - \beta_1^t) / \sqrt{(1 - \beta_2^t)}$. The gradient cancels, so this is a property of the two betas alone.

β_2	step 1	furthest from correct	within 5% of correct from
0.9	$0.32 \times$	$0.32 \times$ at step 1	step 23
0.99	$1.00 \times$	$2.13 \times$ at step 13	step 237
0.999	$3.16 \times$	$6.57 \times$ at step 12	step 2375

At the standard $\beta_2 = 0.999$, an uncorrected Adam takes steps up to **6.6 times** too large, and is still more than 5% wrong at step 2374. Two divisions remove all of it.

PART 7

Adam, one line at a time

Every piece is now built. Rather than write Adam down, change RMSProp's update line one piece at a time and watch Adam appear.

Three lines, and only one thing changes each time

START $w_t = w_{t-1} - \alpha g_t / (\sqrt{v_t} + \epsilon)$ RMSProp, from Part 4

CHANGE 1 $w_t = w_{t-1} - \alpha m_t / (\sqrt{v_t} + \epsilon)$ the averaged direction, not one raw gradient

CHANGE 2 $w_t = w_{t-1} - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$ both averages, corrected

The third line is Adam. Nothing else about the update changed.

Written out in full

ADAM

1 DIRECTION $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$

2 SIZE $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g^2_t$

3 CORRECT $\hat{m}_t = m_t / (1 - \beta_1^t)$ · $\hat{v}_t = v_t / (1 - \beta_2^t)$

4 STEP $w_t = w_{t-1} - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$

Where each line came from

line	what it is	where you met it
1	an average of gradients	Momentum, Worksheet 07
2	an average of squared gradients	RMSProp, Part 4
3	divide each by the weight it has actually applied	Part 6, which you just derived
4	step in the direction of $\hat{m}$, scaled by $\sqrt{\hat{v}}$	AdaGrad's update line, Part 2

There is no new idea on this page. Adam is the three previous parts, assembled.

Step 1, spelled out

The same gradient you have used all sheet: $g_1 = 3$, with $\alpha = 0.1$ and $\beta_1 = \beta_2 = 0.9$. Take the four lines in order.

LINE 1 AND LINE 2 · THE TWO AVERAGES

$m_1 = 0.9(0) + 0.1(3) = \mathbf{0.3000}$ the direction, badly underestimated

$v_1 = 0.9(0) + 0.1(9.0000) = \mathbf{0.9000}$ the size, badly underestimated too

Both are far too small, for the reason you derived in Part 6: m_0 and v_0 were zero and still carry 0.9 of the weight.

LINE 3 · CORRECT BOTH

$1 - 0.9^1 = \mathbf{0.1000}$ the weight actually applied after one step

$\hat{m}_1 = 0.3000 / 0.1000 = \mathbf{3.0000}$ which is g_1

$\hat{v}_1 = 0.9000 / 0.1000 = \mathbf{9.0000}$ which is g^2_1

The correction handed back the gradient itself. At t = 1 the only weight the average applied was $1 - \beta$, and dividing by $1 - \beta$ undoes exactly that.

LINE 4 · THE STEP

$\sqrt{\hat{v}_1} = \sqrt{9.0000} = \mathbf{3.0000}$

$\Delta w = 0.1 \times 3.0000 / 3.0000 = 0.1 \times 1 = \mathbf{0.1000}$

$w_1 = 1.0000 - 0.1000 = \mathbf{0.9000}$

Adam's first step is exactly α , whatever the gradient was

$\hat{m}_1 = (1 - \beta_1)g_1 / (1 - \beta_1) = g_1 \quad \cdot \quad \hat{v}_1 = (1 - \beta_2)g^2_1 / (1 - \beta_2) = g^2_1$

$\hat{m}_1 / \sqrt{\hat{v}_1} = g_1 / |g_1| = \pm 1 \quad \text{so} \quad |\Delta w| = \alpha$

Both betas cancel. Both gradients cancel. Compare RMSProp, whose first step was $\alpha/\sqrt{(1 - \beta_2)}$ and depended on β_2 .

The one sentence worth memorising

$$\text{step} = \alpha \times \hat{m}_t / \sqrt{\hat{v}_t}$$

the numerator, $\hat{m}_t$	where have this parameter's gradients consistently been pointing?
the denominator, $\sqrt{\hat{v}_t}$	how large are this parameter's gradients, usually?

A step is not big because the gradient is big. It is big when the gradient is **bigger than usual for this parameter**.

PART 7 · QUESTION 4

Parameter A's gradients are usually around 100. Parameter B's are usually around 0.01. Both report a gradient of exactly their usual size this step. Adam moves:

- A** A ten thousand times further than B
- B** B further than A, because its gradient is smaller
- C** both about α , because each gradient is ordinary for its own parameter
- D** neither, because $\hat{m}/\sqrt{\hat{v}}$ is zero

What the two betas control

	default	remembers about	why
β_1 , on the direction	0.9	10 gradients	the direction must be able to change when the loss surface turns
β_2 , on the size	0.999	1000 gradients	the size estimate sits in a denominator, so it has to be steady

$\beta_2 > \beta_1$ in almost every published setup. A jumpy denominator makes every step jumpy, which is the failure you saw at $\beta_2 = 0.5$ in Part 4.

The published defaults are $\beta_1 = 0.9, \beta_2 = 0.999, \varepsilon = 10^{-8}$. The next section uses $\beta_2 = 0.9$ so that two steps of hand arithmetic show the correction working. At 0.999 the correction is still doing visible work a thousand steps in.

PART 7 · YOUR TURN

Adam by hand

THE RULE, LEARNING RATE 0.1, BOTH DECAYS 0.9

$m_t = 0.9m_{t-1} + 0.1g_t \cdot v_t = 0.9v_{t-1} + 0.1g^2_t \cdot \hat{m} = m/(1-0.9^t) \cdot \hat{v} = v/(1-0.9^t) \cdot w_t = w_{t-1} - 0.1\hat{m}/\sqrt{\hat{v}}$

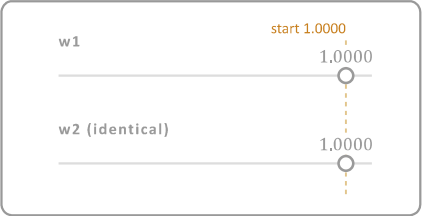
What you are given

$w_1 = 1.0000$

$m_0 = v_0 = 0$

$\alpha = 0.1, \beta_1 = \beta_2 = 0.9$

gradients: 3, then 4 · squared: 9.0000, then 16.0000



STEP 1
from 1.0000
both averages start at zero, so
both need correcting.

YOU NEED $m_0 = v_0 = 0$ $\beta_1 = \beta_2 = 0.9$

$m_1 = 0.9(0) + 0.1(3) = \underline{\hspace{2cm}}$ $v_1 = 0.9(0) + 0.1(9.0000) = \underline{\hspace{2cm}}$

THE AVERAGE OF GRADIENTS

- (a) 0.9000
- (b) 2.7000
- (c) 3.0000
- (d) 0.3000

THE AVERAGE OF SQUARED GRADIENTS

- (a) 9.0000
- (b) 0.9000
- (c) 0.3000
- (d) 8.1000

YOU NEED $m_1 = 0.3000$ $v_1 = 0.9000$

$1 - 0.9^1 = 0.1000$, so $\hat{m}_1 = 0.3000 / 0.1000 = \underline{\hspace{2cm}}$ $\hat{v}_1 = 0.9000 / 0.1000 =$

$\underline{\hspace{2cm}}$

THE CORRECTED $\hat{M}_1$

- (a) 0.3000
- (b) 3.0000
- (c) 0.9000
- (d) 0.0300

THE CORRECTED $\hat{V}_1$

- (a) 3.0000
- (b) 0.0900
- (c) 0.9000
- (d) 9.0000

YOU NEED $\hat{m}_1 = 3.0000$ $\hat{v}_1 = 9.0000$ $\alpha = 0.1$

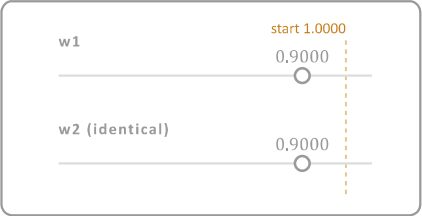
$\sqrt{9.0000} = 3.0000$, so $\Delta w_1 = 0.1 \times 3.0000 / 3.0000 = \underline{\hspace{2cm}}$

- (a) 0.1000
- (b) 0.0316
- (c) 0.3000
- (d) 1.0000

YOU NEED $\text{corrected step} = 0.1000$ $\text{uncorrected step} = 0.0316$

Without the correction the step would have been $0.1(0.3000) \div \sqrt{0.9000} = 0.0316$. The
correction made the first step $\underline{\hspace{2cm}}$

- (a) 3.16 times larger
- (b) 10 times larger
- (c) unchanged
- (d) 3.16 times smaller



STEP 2
from **0.9000**
the correction factor has grown
from 0.1000 to 0.1900.

YOU NEED $m_1 = 0.3000$ $v_1 = 0.9000$ $\beta_1 = \beta_2 = 0.9$

$m_2 = 0.9(0.3000) + 0.1(4) =$ _____ $\cdot v_2 = 0.9(0.9000) + 0.1(16.0000) =$ _____

THE AVERAGE OF GRADIENTS

- (a) 0.2700
- (b) 0.6700
- (c) 3.6000
- (d) 0.4000

THE AVERAGE OF SQUARED GRADIENTS

- (a) 0.8100
- (b) 2.4100
- (c) 1.6000
- (d) 16.9000

YOU NEED $\beta_1 = \beta_2 = 0.9$ $t = 2$

At step 2 the correction factor is $1 - 0.9^2 =$ _____

- (a) 0.8100
- (b) 0.1900
- (c) 0.0100
- (d) 0.1000

YOU NEED $m_2 = 0.6700$ $v_2 = 2.4100$ $1 - 0.9^2 = 0.1900$

$\hat{m}_2 = 0.6700 / 0.1900 =$ _____ $\cdot \hat{v}_2 = 2.4100 / 0.1900 =$ _____

THE CORRECTED $\hat{M}_2$

- (a) 6.7000
- (b) 0.1273
- (c) 0.6700
- (d) 3.5263

THE CORRECTED $\hat{V}_2$

- (a) 2.4100
- (b) 0.4579
- (c) 12.6842
- (d) 24.1000

YOU NEED $\hat{m}_2 = 3.5263$ $\hat{v}_2 = 12.6842$ $\alpha = 0.1$

$\sqrt{12.6842} =$ _____ , then $\Delta w_1 = 0.1(3.5263) \div$ that = _____

THE SQUARE ROOT

- (a) 160.8879
- (b) 6.3421
- (c) 3.5615
- (d) 1.5524

THE STEP FOR w_1

- (a) 0.2577
- (b) 0.0990
- (c) 0.1000
- (d) 0.3526

YOU NEED

w_1 after step 1 = 0.9000

step = 0.0990

$w_1 = 0.9000 - 0.0990 =$ _____

(a) 0.8010

(b) 0.4261

(c) 0.9990

(d) 0.8200

Two properties you can now see in your own numbers

1. The first step was **exactly α** . At $t = 1$ the corrections make $\hat{m}_1 = g_1$ and $\hat{v}_1 = g_1^2$, so the ratio is $g_1/|g_1| = 1$.
2. The second step was **0.0990**, on a gradient of **4**. Plain SGD would have moved $0.1(4) = \mathbf{0.4000}$, four times as far. The ratio $\hat{m}/\sqrt{\hat{v}}$ stays near 1, so an Adam step is about α in size however large the gradient is.

A gradient of 4 is not large in itself. It is large only if 4 is large **for this parameter**, and v says it is not.

TABLE G · ADAM, LEARNING RATE 0.1, BOTH DECAYS 0.9

	step 1	step 2	w after
w_1	0.1000	0.0990	0.8010
w_2	0.1000	0.0990	0.8010
w_1 , correction switched off	0.0316	0.0432	0.9252

The bottom row is the same run with line 3 deleted. It moved 0.0748 in two steps where Adam moved 0.1990.

RETRIEVAL

Match, MCQs, explain

What does each rule keep in memory, per parameter?

AdaGrad → _____

- (a) the sum of every squared gradient since step 1
- (b) an average of squared gradients that forgets
- (c) an average of gradients and an average of squared gradients, both corrected

RMSPProp → _____

- (a) an average of gradients and an average of squared gradients, both corrected
- (b) the sum of every squared gradient since step 1
- (c) an average of squared gradients that forgets

Adam → _____

- (a) an average of squared gradients that forgets
- (b) the sum of every squared gradient since step 1
- (c) an average of gradients and an average of squared gradients, both corrected

RETRIEVAL · QUESTION 5

Why does the rule divide by $\sqrt{G}$ rather than by G ?

- A** To keep the denominator positive.
- B** So the denominator has the same units as the gradient, and the ratio $g/\sqrt{G}$ is a pure number.
- C** To make the arithmetic easier.
- D** Because G grows too slowly otherwise.

RETRIEVAL · QUESTION 6

An AdaGrad run and an RMSPProp run start from the same point with the same α and the same first gradient. Their first steps are:

- A** different: AdaGrad's is α , RMSPProp's is $\alpha/\sqrt{1 - \beta_2}$
- B** identical, because v_1 and G_1 are both built from g^2_1
- C** different: AdaGrad's is larger
- D** impossible to say without knowing the gradient

RETRIEVAL · QUESTION 7

You set $\beta_2 = 0.5$ in RMSPProp. Compared with $\beta_2 = 0.99$, the denominator will:

- A** grow without limit, like AdaGrad's
- B** stay closer to constant
- C** track recent gradients closely and jump around more
- D** have no effect on the step size

RETRIEVAL · QUESTION 8

Adam stores two running averages per parameter. Against plain SGD, that costs:

A

nothing, they are scalars

B

one extra copy of the parameters

C

about 50% more memory

D

two extra copies of the parameters

RETRIEVAL · QUESTION 9

In the numerical you worked, Adam's first step was exactly $\alpha = 0.1000$. That happens:

A

only because the gradient was 3

B

for any first gradient, because both corrections divide by the same factor at $t = 1$

C

only because $\beta_1 = \beta_2$

D

only when $v_0 = 0$

The four rules, in one place

TABLE H · THE FOUR RULES, SIDE BY SIDE

rule	what it stores per parameter	how it fails
SGD	nothing	one α for parameters that need different step sizes
Momentum	an average of gradients	fixes the direction, not the distance
AdaGrad	G , the sum of squared gradients	G only grows, so the step decays to zero
RMSProp	v , an average of squared gradients	starts at zero, so the first steps are too large; no direction memory
Adam	m and v , both corrected	two more hyperparameters, and twice the memory of SGD

Explain, in your own words

- Write AdaGrad's two lines from memory. Say what G is and why it is squared.
- Explain, without arithmetic, why a parameter with large gradients ends up with a small step.
- State the one line that separates RMSProp from AdaGrad, and the failure it removes.
- AdaGrad's first step is always α . RMSProp's is $\alpha/\sqrt{(1 - \beta_2)}$. Explain the difference from the two accumulator rules.
- Say what would go wrong if Adam corrected m but not v .
- In Part 1 momentum brought 574 steps down to 539. Explain why an adaptive rule can do so much better than that.

Every value is rounded to 4 decimals at every step, so a calculator reproduces this sheet exactly.

Reference: S. J. D. Prince, *Understanding Deep Learning*, Section 6.4. Original papers: Duchi et al. 2011 (AdaGrad), Tieleman & Hinton 2012 (RMSProp), Kingma & Ba 2015 (Adam).